Task Report - Day 2

Title

TASK-02: Developing a Rule-Based NLP Engine for Live News Streams

Objective

The objective of Task 2 is to connect to the News API, fetch current news articles based on a keyword, and build a simple rule-based news processing system. The script is designed to classify each news headline into categories such as FINANCE, TECH, POLITICS, or GENERAL, and also extract useful entities like company names, percentages, and currency values from the news description. The final processed data is then stored in a structured JSON file for further analysis.

Libraries Used

1. **requests**
    - Used to send HTTP requests to the News API and fetch live news data.
2. **json**
    - Used to read the category configuration file and write the final processed output into output.json.
3. **os**
    - Used to handle file paths and access the .env file location.
4. **re**
    - Used for regular expression matching to extract entities such as percentages and currency values.
5. **time**
    - Used to handle rate-limiting by adding retry delays when the API request fails or times out.

Errors

1. **Incorrect Output File Name**

    - **Issue:** The script was saving data in jsonoutput.json instead of output.json.
    - **Solution:** The output file path was corrected so the script always saves data into output.json.

2. **Missing dotenv Module**

    - **Issue:** The script raised ModuleNotFoundError: No module named 'dotenv'.
    - **Solution:** The environment variable handling was simplified by either installing python-dotenv or manually reading the .env file in the code.

3. **Using the Wrong URL**

    - **Issue:** The website homepage URL https://newsdata.io/ was used instead of the actual API endpoint.
    - **Solution:** The URL was corrected to the proper API endpoint: https://newsdata.io/api/1/latest.

4. **Incorrect Company Name Extraction**

    - **Issue:** The initial regex extracted all capitalized words as company names.
    - **Solution:** Company detection was improved by defining company names in category_config.json and matching only those names in the article headline and description.

5. **Duplicate News Articles in Output**

    - **Issue:** The same article headline appeared multiple times in the output JSON.
    - **Solution:** A duplicate-checking step was added using a seen_headlines set so that repeated headlines are skipped before saving.